

Anime Character Colorization using Few-shot Learning

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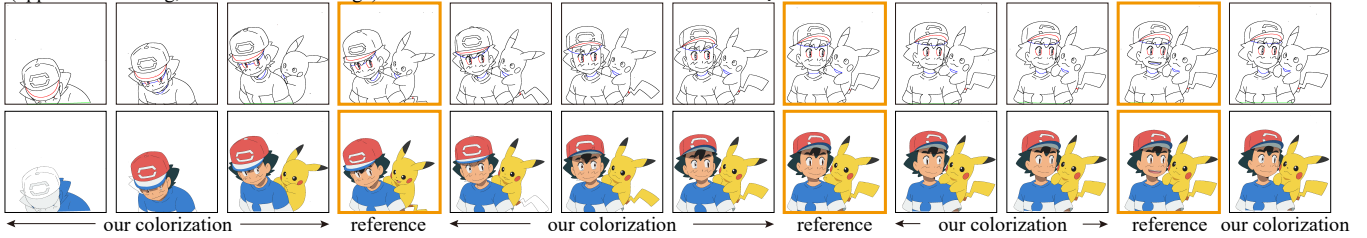


Figure 1: Anime character colorization for line-drawings using a few reference images (surrounding yellow boxes). Note, lines on the upper rows are dilated for visibility.

ABSTRACT

In this paper, we propose an automatic Anime-style colorization method using only a small number of colored reference images manually colorized by artists. To accomplish this, we introduce a few-shot patch-based learning method considering the characteristics of Anime line-drawing. To streamline the learning process, we derive optimal settings with acceptable colorization accuracy and training time for a production pipeline. We demonstrate that the proposed method helps to reduce manual labor for artists.

CCS CONCEPTS

• Computing methodologies → Image manipulation.

KEYWORDS

Anime character, colorization, few-shot learning

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1 INTRODUCTION

In the Anime-production pipeline, colorization process of the draft line-drawings remains a tedious and time-consuming task even if the pipeline is digitized. The process consists in manually filling the regions surrounded by contour lines with the colors specified in the color script of the character.

With the evolution of recent deep learning technologies, deep learning-based colorization methods have been proposed by [Furusawa et al. 2017; Ishii et al. 2020; Ramassamy et al. 2018]. These methods, in general, require a lot of learning-ready image pairs, which consist of line-drawing and colored images with/without semantic region label, for target characters.

However, sufficient number of image pairs is not always available for minor characters who appear rarely in the footage. The few-shot learning approach is promising for such cases. Recently, Texler et al. [2020] proposed a patch-based learning method using a small number of reference images and applied it to interactive style (texture) transformation of images. We adopt this method for colorizing Anime character's line-drawing only using few reference images. However, Anime-style drawings are different from natural images used in their study [2020]. The line-drawing images contain insufficient clues for automatic colorization.

We overcome this issue with the following contributions:

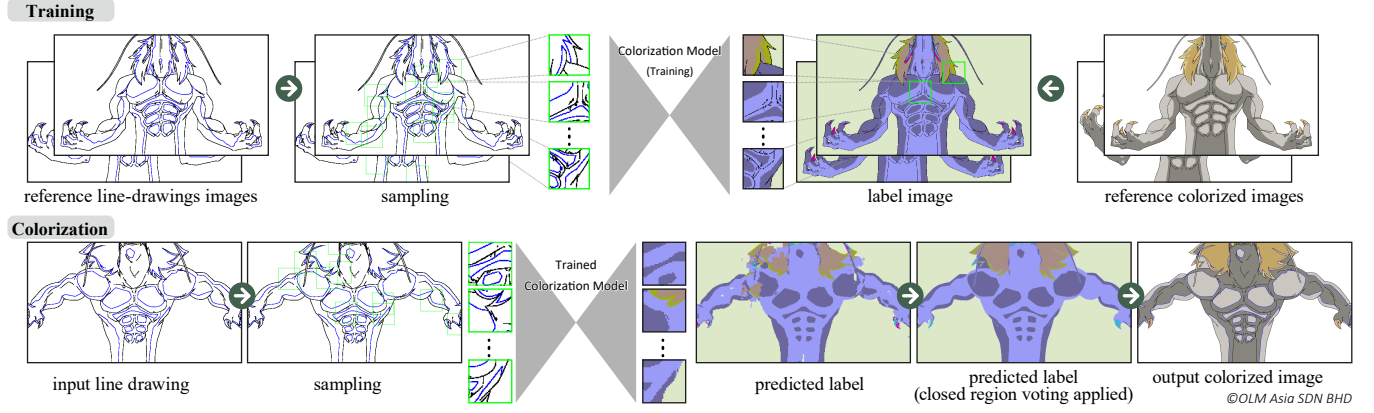


Figure 2: Overview of the proposed method

- (1) Introduce a simple but effective patch sampling method specific to Anime line-drawings.
- (2) Concatenate spatial information onto the deep features to help colorization of an input image sequence in which the layout is loosely consistent.
- (3) Derive optimal patch size for a good balance of colorization accuracy and training time.

2 RELATED WORK

2.1 Colorization without reference images

There are several deep learning based automatic colorization methods for natural gray-scale images [Iizuka et al. 2016; Zhang et al. 2017], comic illustration [Ci et al. 2018; Hensman and Aizawa 2017]. Generated colors of these methods tend to have color gradations which look like water color painting style. On the other hand, Anime colorization has a unique style that uses uniform colors painted over closed regions, and only colors specified by a color director may be used. Therefore, the above mentioned methods are not suitable for our purpose. Ishii et al. [2020] proposed an automatic colorization method for Anime characters [Ishii et al. 2020]. Their method satisfies the style of Anime colorization. However, every target character requires several learning ready image pairs (line-drawing and manually colorized images). It is not easy to collect those images because additional tasks including noise clean up, gap filling for contour lines and removing inconsistent pairs are required during/after production. Moreover, minor characters rarely appear in the footage. Therefore, we adopt a few-shot learning strategy that makes an automatic colorization method more acceptable in a production pipeline.

2.2 Colorization with reference images

Automatic colorization using reference images has also been widely studied, including methods for natural gray-scale images [Gupta et al. 2012; Liu et al. 2008], rough sketch line drawings [Lee et al. 2020]. Kanamori [2012]; Sato et al. [2014] addressed Anime-style colorization as a class of region matching problems between reference images and target images. Those methods are based on region matching and are not robust enough to topology changes.

3 METHOD

3.1 Overview

We address the colorization problem as a semantic segmentation one as stated in [Ishii et al. 2020]. In this paper, we propose a colorization method that relies on a few-shot learning approach. The overview of the proposed method is shown in Figure 2. Our dataset is composed of pairs on a few-shot reference colorized images and corresponding line-drawing images. The user just needs to feed a small number of pairs to the network for the training. We first construct a color palette which maps label IDs to colors from a set of reference colorized images. Then, the reference color images are encoded to label images using the color palette. Using the pairs of line-drawing images and corresponding label images, we train our colorization model as described in Section 3.2.

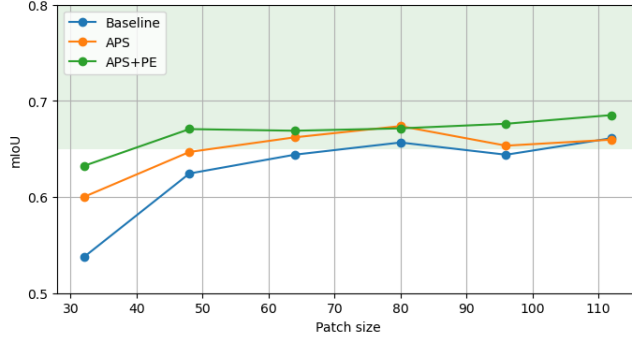
The colorization process (as shown in Figure 2-Colorization) predicts the labels from an input line drawing. This process is almost the same as the training one. After the label prediction, we just add a label voting step as presented in [Ramassamy et al. 2018]. Finally, according to the color palette, the label IDs of the resulting images are decoded to colors.

3.2 Anime-specific Patch-based Learning

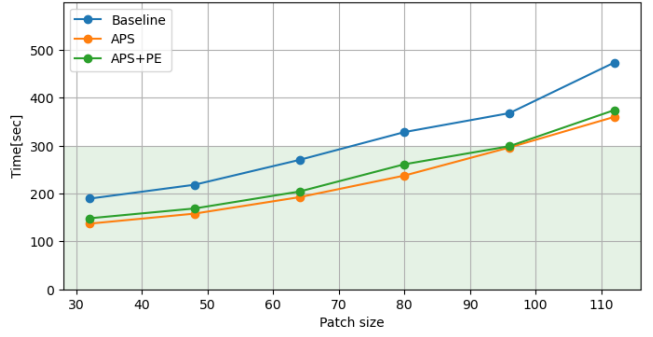
We build our colorization model on few-shot patch-based learning for natural image style transfer [Texler et al. 2020]. To address Anime-style colorization as the semantic segmentation problem, we make the generator output label IDs and replace the L1 loss function of the generator with the cross entropy loss function. Hereafter, we call this customized method: “Baseline”.

The simple random sampling used in the Baseline method as described in [Texler et al. 2020] is inefficient. The line drawing images are intrinsically sparse so a random sampled patch does not always contain sufficient variations. Anime-style images have the characteristic that the color change occurs at boundary or junction with lines. To satisfy the sampling variation, we need to ensure that the patch contains lines, i.e. color variation.

Thus, we propose a simple but effective sampling method. We apply corner detection proposed by Shi and Tomasi [1994] to the line-drawings. Then, points are randomly selected based on the



(a) Colorization Accuracy (mIoU)



(b) Training Time

Figure 3: Comparison between Baseline, APS and APS+PE with respect to colorization accuracy and training time. Each green span represents the requirement for accuracy and training time.



Figure 4: Colorization result with different patch size. One of the actual predicted patches is shown in upper left for each images.

detected corners, within the limit to the number of patches, specified by users. We called this sampling method *Anime-specific Patch Sampling (APS)*. Patches are extracted by cutting out square regions around the points from reference line-drawing images and colorized images.

In patch sampling, similar line-drawing patches may correspond to different labeled patches. This may lead to one-to-many mapping which would makes learning difficult. To prevent this mapping, we embed spatial information into sampled patches to make line-drawing patches unique. We start by obtaining the gravity center of all sampled points in image coordinates, then compute the position of all the pixels in all the squares to be cut out as patches, relative to the gravity center. Finally, we embed the resulting relative 2D coordinates into each patch as two additional channels. As a result, this *Positional Encoding (PE)* maintains consistent predictions between similar patches sampled at different locations.

In patch-based learning, patch size is an important factor to determine accuracy and training time. Since it has a big impact on resulting accuracy, we discuss the optimal patch size for our model in Section 4.2.

4 EVALUATION EXPERIMENT

We created a test set for the evaluation experiment by collecting 9 sets of line drawings and hand-colored images from 3 Anime works of different styles. The test set consists of a number of line drawings (average of 12), reference pairs of line-drawing images and corresponding colorized images (average of 4). The resolution

of the images in the test set ranges from 680×440 pixels to 704×496 pixels but our method does not depend on the resolution of the input image. All experiments in this paper were performed on Ubuntu 18.04, Intel Xeon 3.6 GHz, NVIDIA Quadro RTX 8000.

4.1 Evaluation Criteria

We use mean the Intersection-over-Union (mIoU) metric commonly used in the evaluation of semantic segmentation as a colorization accuracy criterion, with

$$\text{mIoU} = \frac{1}{K} \sum_{i=1}^K \frac{N_{ii}}{\sum_{j=1}^K (N_{ij} + N_{ji}) - N_{ii}}, \quad (1)$$

where i, j are the label IDs, N_{ij} is the number of pixels that the generator classified a pixel with class i in ground truth as class j . K is the number of labels excluding the label corresponding to the color of the contour line in the color script. If mIoU is close to 1, it shows that the result is close to the correct manual coloring result.

4.2 Accuracy and Training Time

For the Baseline, APS and APS+PE methods, we first measured the average of both colorization accuracy and training time with respect to patch size of values; 32, 48, 64, 80, 96 and 112 (=approximately a fourth of the input image height) using the test set. In all measurements, the batch size and the number of mini-batches used for training are fixed to 16 and 1,000 respectively. The results are shown in Figure 3. Our method (APS and APS+PE) converges

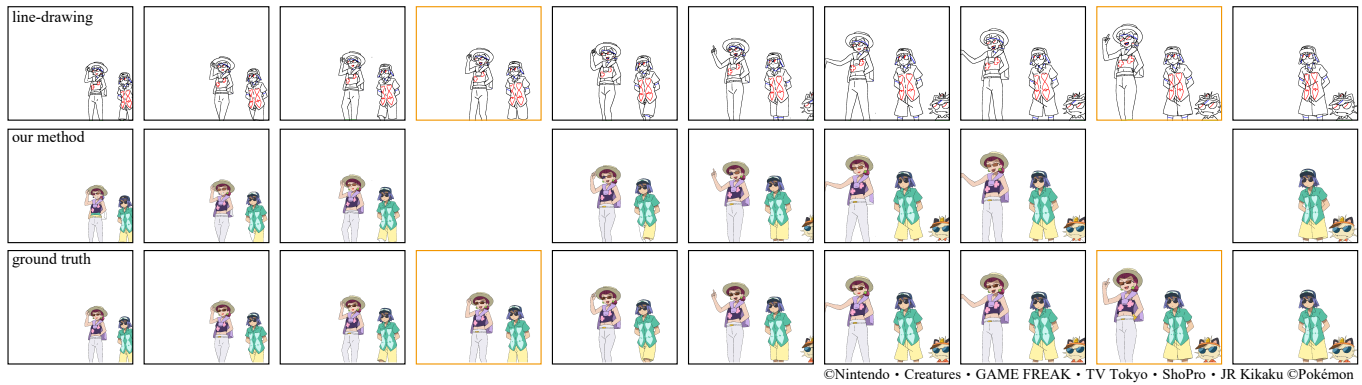


Figure 5: Another colorization result using APS+PE with patch size 64. Reference images are surrounded by yellow boxes. Note, lines on the upper row are dilated for visibility.

faster than the Baseline method with shorter training time and better accuracy. It means that our APS acquires useful patches more efficiently than random sampling. The PE proves to improve even more the convergence speed. We can then use quite small patches (around 32–64 pixels in our test) and guarantee minimal training time (approximately 200 s) as shown in Figure 3.

Figure 4 shows colorization results using APS+PE with different patch sizes. It clearly shows that using larger patches helps to improve the colorization accuracy but more training time is required as shown in Figure 3. There is a trade-off between the patch size and training time. The mIoU of APS+PE seems to converge with a patch size of 64. The appearance of colorization results does not change when using patches larger than 64. We could achieve enough accuracy with patches of size 64 and the 1,000 training cycles. This requires around approximately 200 s of training. We show colorization results in Figures 1 and 5.

5 USER STUDY AND CONCLUSION

To evaluate our automatic colorization, we asked professional colorization artists to use the proposed system. The interviews after the experiment gave us the following insights:

- In the case of images with simple structures, considering the time required to check and correct painting errors, automatic colorization needs to be completed within 5–10 min for 20–30 images and we do it in approximately 3 min 20 s, so that’s acceptable.
- Results with $mIoU \geq 0.65$ give a good starting point for manual colorization and reduce manual labor time.
- The more complex the structure of an image, the more time reduction of the manual labor time using our colorization system.

We highlighted these artist requirements in Figure 3 with green bands. So training with a patch size between 48 and 64 pixels in 200 s seemed to be acceptable for our artists. We therefore conclude that we developed a system that satisfies production needs.

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